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ODD END SEMESTER EXAMINATION DECEMBER- 2024

Name of the Course: B.Tech (ME)
 Name of the paper: Thermal Engineering
 Time: 3 hours

Semester: III
 Paper Code: TME 308
 Maximum Marks: 100

Note:-

- (i) All questions are compulsory
- (ii) Answer any two sub questions among a, b, & c in each main question
- (iii) Total marks in each main question are twenty
- (iv) Each question carries 10 marks

Q1. 20 marks	
(a) State the first law of thermodynamics for a closed system undergoing a cycle as well as a change of state and explain its application.	CO1
(b) A gas is compressed in a cylinder from a volume of 0.8 m^3 to 0.4 m^3 . The pressure-volume relationship for the process is given by $PV^{1.4} = \text{constant}$. If the initial pressure is 300 kPa, determine the work done on the gas during the compression.	
(c) Steam initially at 1.5 MPa, 300°C expands reversibly and adiabatically in a steam turbine to 40°C . Determine the ideal work output of the turbine per kg of steam.	
Q2. 20 marks	
(a) What is a heat pump? How does it differ from a refrigerator? Show that the COP of a heat pump is greater than the COP of a refrigerator by unity.	CO2
(b) In a steady flow apparatus, 13.5 kJ of work is done by each kg of fluid. The specific volume of the fluid, pressure, and velocity at the inlet are $0.37 \text{ m}^3/\text{kg}$, 600 kPa, and 16 m/s. The inlet is 32 m above the floor, and the discharge pipe is at floor level. The discharge conditions are $0.62 \text{ m}^3/\text{kg}$, 100 kPa, and 270 m/s. The total heat loss between the inlet and discharge is 9 kJ/kg of fluid. In flowing through this apparatus, does the specific internal energy increase or decrease, and by how much?	
(c) Two Carnot engines work in series between the source and sink temperatures of 550 K and 350 K. If both engines develop equal power, determine the intermediate temperature.	
Q3. 20 marks	
(a) Explain the principle of increase of entropy. Derive an expression for the change in entropy for isobaric and isochoric process.	CO3, CO4
(b) Show that the internal energy and enthalpy of an ideal gas are functions of temperature only.	
(c) Explain and derive Maxwell's equations.	

Q4.	20 marks	
(a)	Explain the working principle of Otto cycle with the help of p-v and T-s diagram and also derive the air standard efficiency of the cycle.	CO5
(b)	Consider a steam power plant operating on the simple ideal Rankine cycle. Steam enters the turbine at 3 MPa and 350°C and is condensed in the condenser at a pressure of 75 kPa. Determine the thermal efficiency of this cycle.	
(c)	A gas-turbine power plant operating on an ideal Brayton cycle has a pressure ratio of 8. The gas temperature is 300 K at the compressor inlet and 1300 K at the turbine inlet. Utilizing the air-standard assumptions, determine (i) the gas temperature at the exits of the compressor and the turbine, (ii) the back work ratio, and (iii) the thermal efficiency.	
Q5.	20 marks	
(a)	Explain, with the help of a neat sketch, the principle of operation of a single stage, single acting reciprocating compressor.	CO6
(b)	Derive an expression for clearance volumetric efficiency of compressor and simplify it for isentropic process.	
(c)	A refrigerator uses refrigerant-134a as the working fluid and operates on an ideal vapor-compression refrigeration cycle between 0.14 and 0.8 MPa. If the mass flow rate of the refrigerant is 0.05 kg/s, determine (i) the rate of heat removal from the refrigerated space and the power input to the compressor, (ii) the rate of heat rejection to the environment, and (iii) the COP of the refrigerator.	